

Almost-idempotent positive maps and approximate Jordan algebras

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(collaborative working report)

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Abstract

Kitaev showed that a quantum channel which is almost idempotent — a unital completely positive map Φ on the operators of a finite-dimensional Hilbert space with $\|\Phi^2 - \Phi\|_{\text{cb}} \leq \eta$ — carries, on its near-fixed points, the structure of an approximate C^* -algebra, and that every finite-dimensional approximate C^* -algebra is close to a genuine one. We study the same question when Φ is only assumed *positive*, not completely positive. We show that the correct invariant is then a *Jordan* algebra rather than an associative one, for a precise structural reason: a positive unital map need not satisfy the Schwarz inequality $\Phi(x)^*\Phi(x) \leq \Phi(x^*x)$, but it always satisfies Kadison’s inequality $\Phi(x)^2 \leq \Phi(x^2)$ for self-adjoint x , and the latter involves exactly the data of a Jordan algebra.

Our main complete result is a *bridge theorem*: for any unital positive Φ with $\|\Phi^2 - \Phi\| \leq \eta$, the near-fixed-point space $A = \text{Im } P$ of the spectral idempotent $P = \theta(2\Phi - 1)$, equipped with the product $a \bullet b = P(a \circ b)$, satisfies the axioms of a Jordan algebra up to an error $O(\sqrt{\eta})$. The proof uses no dilation and no complete positivity; the key fact is that the “square holes” $P(a^2) - a^2$ are almost-positive null elements, so that quadratic error terms are $O(\eta)$ while the exact Jordan identity of the ambient algebra cancels the leading term. We then formulate the structure-theory programme that would complete the analogue of Kitaev’s theorem — every finite-dimensional approximate Jordan algebra is close to a genuine one — and report honestly on what is proved and what remains open, including the exponent (1/2 in general, 1 conjecturally for decomposable maps) and the dimension-free error-reduction step.

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1 Introduction

The question. A physical system sometimes hides a smaller logical system inside it: the logical observables are encoded in the physical ones, and a recovery operation returns the system to the encoded subspace. Mathematically, such a recovery is a quantum channel T that is idempotent, $T^2 = T$. Kitaev [8] studied the realistic case where the recovery is only *approximately* idempotent, and asked what algebraic structure the encoded observables carry. He worked with the dual map Φ of the channel, a unital completely positive map on the algebra $B(\mathcal{H})$ of operators on a finite-dimensional Hilbert space, and measured non-idempotence by $\|\Phi^2 - \Phi\|_{\text{cb}} \leq \eta$. His answer: the near-fixed points of Φ form an *approximate C^* -algebra*, and every finite-dimensional approximate C^* -algebra is, with universal (dimension-free) constants, close to a genuine C^* -algebra.

This report. We ask the same question for maps that are only *positive*, not completely positive. This is the natural setting whenever one cares about distinguishability of states rather than full quantum operations, and it is the setting of van Luijk and Wilming [11], who show that for positive trace-preserving maps the relevant invariant is a *Jordan algebra*. The present report does three things, in order of completeness:

1. It records the consensus *definitions* (positive map, Jordan product, JB-algebra, order-unit space, approximate Jordan algebra). Standard background definitions are matched against local source copies; project definitions such as the ε -JB notion are marked (CONSENSUS) and tied to internal proof files in PROVENANCE.md.
2. It proves the *bridge theorem* (Section 6): from a unital positive almost-idempotent Φ one obtains, with no use of dilations or complete positivity, a space that satisfies the Jordan-algebra axioms up to $O(\sqrt{\eta})$. This is the positive-map analogue of the first half of Kitaev’s construction, and it is complete.
3. It states the *structure-theory programme* (Section 7) — the analogue of the second half of Kitaev’s theorem — and reports precisely what is established and what is still open (Section 8).

Why Jordan, and not associative. The reason positivity forces a Jordan, rather than associative, structure can be stated in one line. Kitaev’s verification of the C^* -norm axiom rests on the Schwarz inequality $\Phi(x)^*\Phi(x) \leq \Phi(x^*x)$, which holds for 2-positive (in particular completely positive) maps. A merely positive unital map need not satisfy it; what it does satisfy is Kadison’s inequality $\Phi(x)^2 \leq \Phi(x^2)$ for self-adjoint x [7]. The latter involves only self-adjoint elements and squares — exactly the data on which a Jordan algebra is built. Replacing the associative product

xy by the symmetric product $x \circ y = \frac{1}{2}(xy + yx)$ throughout is therefore not a convenience but a necessity, and the entire theory reorganises around it.

Conventions and status flags. All Hilbert spaces are finite-dimensional and complex. We write $B(\mathcal{H})$ for the operators on $\mathcal{H}$ and $B(\mathcal{H})_{\text{sa}}$ for the self-adjoint ones. Statements carry a status: (PROVED) for results with a complete proof in this report; (CITED) for results transcribed verbatim from a named source; (CONSENSUS) for definitions and conventions adopted by agreement between the authors (a definition is never “proved”); and (OPEN) for targets of the programme not yet established. Constants denoted C are universal and never depend on $\dim \mathcal{H}$ unless explicitly stated.

2 Preliminaries

This section fixes the analytic setting. Everything is finite-dimensional; the only norm is the operator norm, and the only positivity used is positivity of maps (never complete positivity).

Definition 2.1 (Operators, self-adjoint part, states). (CITED) Let $\mathcal{H}$ be a finite-dimensional complex Hilbert space and $B(\mathcal{H})$ the algebra of linear operators on $\mathcal{H}$, with the operator norm $\|x\| = \sup_{\|\xi\| \leq 1} \|x\xi\|$. The *self-adjoint part* is $B(\mathcal{H})_{\text{sa}} = \{x \in B(\mathcal{H}) : x = x^*\}$, a real vector space. For $a \in B(\mathcal{H})_{\text{sa}}$ we write $a \geq 0$ if every eigenvalue of a is nonnegative; this induces the partial order $a \geq b \iff a - b \geq 0$, with unit $\mathbf{1}$ the identity operator. A *state* is a linear functional $\omega : B(\mathcal{H}) \rightarrow \mathbb{C}$ that is positive ($\omega(a) \geq 0$ for $a \geq 0$) and unital ($\omega(\mathbf{1}) = 1$).

Definition 2.2 (Positive, unital, and positive unital maps). (CITED) A linear map $\Phi : B(\mathcal{H}) \rightarrow B(\mathcal{H})$ is *positive* if $\Phi(a) \geq 0$ whenever $a \geq 0$; a positive map is automatically self-adjointness-preserving, $\Phi(a^*) = \Phi(a)^*$, so it restricts to a real-linear map $\Phi : B(\mathcal{H})_{\text{sa}} \rightarrow B(\mathcal{H})_{\text{sa}}$. It is *unital* if $\Phi(\mathbf{1}) = \mathbf{1}$. A *positive unital map* is a contraction on the self-adjoint part: for $a \in B(\mathcal{H})_{\text{sa}}$ one has $-\|a\| \mathbf{1} \leq a \leq \|a\| \mathbf{1}$, hence $-\|a\| \mathbf{1} \leq \Phi(a) \leq \|a\| \mathbf{1}$ by positivity and unitality, so $\|\Phi(a)\| \leq \|a\|$; with $\Phi(\mathbf{1}) = \mathbf{1}$ this gives $\|\Phi\| = 1$ on $B(\mathcal{H})_{\text{sa}}$.

The order structure of $B(\mathcal{H})_{\text{sa}}$ is what survives when the associative product is dropped; it is captured abstractly by the notion of an order-unit space.

Definition 2.3 (Order-unit space and order-unit norm). (CITED) A real vector space A with a proper convex cone A^+ (giving $a \geq b \iff a - b \in A^+$) and a distinguished element $e \in A^+$ is an *order-unit space* if (i) e is an *order unit*: for each $a \in A$ there is $\lambda > 0$ with $-\lambda e \leq a \leq \lambda e$, and (ii) A is *Archimedean*: $na \leq e$ for all $n \in \mathbb{N}$ implies $a \leq 0$. The *order-unit norm* is

$$\|a\| = \inf\{t > 0 : -te \leq a \leq te\}.$$

With $e = \mathbf{1}$ and cone $A^+ = \{a \in B(\mathcal{H})_{\text{sa}} : a \geq 0\}$, the self-adjoint part $B(\mathcal{H})_{\text{sa}}$ is an order-unit space, and its order-unit norm coincides with the operator norm.

Remark 2.4. (PROVED) That $B(\mathcal{H})_{\text{sa}}$ is an order-unit space is immediate from Definition 2.3: for self-adjoint a , the spectral bound $-\|a\| \mathbf{1} \leq a \leq \|a\| \mathbf{1}$ shows $\mathbf{1}$ is an order unit and that the order-unit norm equals the spectral-radius (operator) norm; the Archimedean property holds because $B(\mathcal{H})_{\text{sa}}$ is finite-dimensional. In the channel setting of this report the order-unit structure of the relevant subspace is inherited *exactly* from $B(\mathcal{H})_{\text{sa}}$; only the product will be approximate.

The single positivity inequality available for merely positive unital maps is Kadison’s, with its symmetrized strengthening due to Størmer.

Proposition 2.5 (Kadison and Jordan–Schwarz inequalities). (CITED) *Let $\Phi : B(\mathcal{H}) \rightarrow B(\mathcal{H})$ be a positive unital map.*

(i) (Kadison [7]) *For every self-adjoint $a \in B(\mathcal{H})_{\text{sa}}$,*

$$\Phi(a)^2 \leq \Phi(a^2).$$

(ii) (Jordan–Schwarz, Størmer [10, 11]) *For every $a \in B(\mathcal{H})$, writing $\circ$ for the Jordan product $x \circ y = \frac{1}{2}(xy + yx)$,*

$$\Phi(a^*) \circ \Phi(a) \leq \Phi(a^* \circ a).$$

Part (i) is the case $a = a^$ of part (ii). A merely positive unital map need not satisfy the Schwarz inequality $\Phi(a)^* \Phi(a) \leq \Phi(a^*a)$, which requires 2-positivity.*

Remark 2.6 (Operator norm in place of the cb-norm). (PROVED) (*framing; original to this report*) Kitaev [8] measures non-idempotence in the completely bounded norm, $\|\Phi^2 - \Phi\|_{\text{cb}} \leq \eta$, which presupposes that the relevant maps are completely bounded. We instead use the operator norm of the map on $B(\mathcal{H})_{\text{sa}}$, $\|\Phi^2 - \Phi\| \leq \eta$. The reason is structural: mere positivity is not stable under matrix amplification: $\text{id}_n \otimes \Phi$ need not be positive. In finite dimension every linear map is bounded (and completely bounded once an operator-space structure is fixed), so the obstruction is not existence of a cb-norm; it is that the cb/tensor norm is not the order-norm control supplied by positivity alone. The cb-norm and complete positivity enter Kitaev’s argument only through the dilation/tensor machinery (Stinespring, Choi); the operator-norm formulation keeps every estimate intrinsic to $B(\mathcal{H})_{\text{sa}}$ and dilation-free, at the cost of the weaker exponent recorded in the bridge theorem (Section 6). Throughout, the spectral idempotent is denoted P and the projected Jordan product is written $a \bullet b := P(a \circ b)$.

Agent B review correction. The earlier draft said that a positive map need not be completely bounded. In the finite-dimensional setting of this report that sentence is false; the correct issue is failure of positivity under amplification.

3 Jordan algebras and JB-algebras

This section fixes the algebraic vocabulary used throughout the report. Every object here is standard; we transcribe the axioms verbatim from Hanche-Olsen and Størmer [5] so that later approximate versions (Section 5) can be read as quantitative relaxations of these exact identities. We work over a field of characteristic not 2; for the analytic results the scalars are real, as indicated.

3.1 Jordan algebras

The Jordan product symmetrises an associative product: for operators on a Hilbert space, $a \circ b = \frac{1}{2}(ab + ba)$ is self-adjoint whenever a and b are, so it keeps us inside $B(\mathcal{H})_{\text{sa}}$ where the associative product ab does not. Abstracting the two identities that this symmetrised product satisfies gives the notion of a Jordan algebra.

Definition 3.1 (Jordan algebra). (CITED) For elements a, b of an associative algebra, the *Jordan product* is

$$a \circ b = \frac{1}{2}(ab + ba). \tag{1}$$

An algebra A with a product written $(a, b) \mapsto a \circ b$ is called a *Jordan algebra* if the following two identities hold for all $a, b \in A$:

$$a \circ b = b \circ a, \quad (2)$$

$$a \circ (b \circ a^2) = (a \circ b) \circ a^2. \quad (3)$$

Identity (2) states that A is commutative; (3) is the *Jordan identity*. Here $a^2 := a \circ a$.

Remark 3.2 (Form of the Jordan identity). (CITED) The Jordan identity (3), written $a \circ (b \circ a^2) = (a \circ b) \circ a^2$ in [5], is the same as the symmetric-looking form

$$((a \circ a) \circ b) \circ a = (a \circ a) \circ (b \circ a),$$

obtained by writing $a^2 = a \circ a$ and applying commutativity (2) to each product. We use whichever form is more convenient; they are equivalent in any commutative algebra.

A Jordan algebra need not be associative. Its one indispensable substitute for associativity is that any single element, together with its powers, generates an associative subalgebra. This is what makes a functional calculus possible later.

Proposition 3.3 (Power-associativity). (CITED) *Let A be a Jordan algebra and $a \in A$. Define powers inductively by $a^1 = a$ and $a^{n+1} = a \circ a^n$ (and $a^0 = \mathbf{1}$ if A is unital). Then the subalgebra generated by a is associative; equivalently, for all natural numbers m, n ,*

$$a^{m+n} = a^m \circ a^n. \quad (4)$$

Consequently every power a^n is unambiguously defined.

3.2 JB-algebras

To do analysis one adds a complete norm compatible with the product. The two extra norm conditions below are the Jordan analogue of the C^* -identity: the first fixes the norm of a square, the second forces sums of squares to dominate their summands and is what makes the order structure (positivity = being a square) well-behaved.

Definition 3.4 (JB-algebra). (CITED) A *Jordan Banach algebra* is a real Jordan algebra A equipped with a complete norm satisfying

$$\|a \circ b\| \leq \|a\| \|b\|, \quad a, b \in A. \quad (5)$$

A *JB-algebra* is a Jordan Banach algebra A whose norm satisfies, in addition, the following two conditions for all $a, b \in A$:

$$\|a^2\| = \|a\|^2, \quad (6)$$

$$\|a^2\| \leq \|a^2 + b^2\|. \quad (7)$$

If A is unital with unit $\mathbf{1}$, then (6) gives $\|\mathbf{1}\| = 1$.

The motivating example is concrete: for a complex Hilbert space $\mathcal{H}$, the self-adjoint operators $B(\mathcal{H})_{\text{sa}}$ form a JB-algebra under (1) and the operator norm. A norm-closed Jordan subalgebra of some $B(\mathcal{H})_{\text{sa}}$ is called a *JC-algebra* [5, 3.1.2], and every JC-algebra is a JB-algebra. The content of the finite-dimensional theory is that, with one exception, the converse also holds.

3.3 Finite-dimensional classification

In finite dimensions the JB-algebras are exactly the classical objects of Jordan, von Neumann and Wigner [6]: a real Jordan algebra A is called *formally real* (or Euclidean) if $\sum_{i=1}^n a_i^2 = 0$ in A forces $a_1 = \dots = a_n = 0$. A finite-dimensional unital formally real Jordan algebra is a JB-algebra in its order norm [5, 3.1.7], and conversely every JB-algebra is formally real [5, 3.3.8]; so in finite dimensions the two classes coincide. Their simple pieces are completely classified.

Theorem 3.5 (Jordan–von Neumann–Wigner classification). (CITED) *Every finite-dimensional, formally real, unital Jordan algebra is a direct sum of simple ones. Each simple summand is isomorphic to exactly one of*

- the one-dimensional algebra $\mathbb{R}$;
- a spin factor $\mathbb{R}\mathbf{1} \oplus H$, where H is a real inner-product space of dimension ≥ 2 (the case of two minimal projections summing to $\mathbf{1}$);
- a Hermitian matrix algebra $H_n(\mathbb{R})$, $H_n(\mathbb{C})$, or $H_n(\mathbb{H})$ with $n \geq 3$, where $\mathbb{H}$ denotes the quaternions; or
- the Albert algebra $H_3(\mathbb{O})$ of 3×3 Hermitian matrices over the octonions $\mathbb{O}$.

Of these, only $H_3(\mathbb{O})$ is exceptional: it is not isomorphic to a Jordan subalgebra of self-adjoint operators on any Hilbert space; all the others are JC-algebras.

Agent B review correction. The one-dimensional summand $\mathbb{R}$ is included explicitly; it is the $n = 1$ case in Hanche-Olsen–Størmer Theorem 2.9.6 and is not covered by the listed spin factors.

Remark 3.6 (Status of the exceptional algebra). (CITED) The exceptionality of $H_3(\mathbb{O})$ traces back to the non-associativity of the octonions [5, 2.8.5]. It is the only obstruction to representing a finite-dimensional formally real Jordan algebra by self-adjoint operators, and it is the reason the structure theory of Section 7 must allow for a Jordan, rather than associative, limit object.

4 The spectral idempotent

Throughout this section $\Phi: B(\mathcal{H})_{\text{sa}} \rightarrow B(\mathcal{H})_{\text{sa}}$ is a fixed unital positive linear map that is *almost idempotent*: there is a parameter $\eta \in [0, \frac{1}{4})$ with

$$\|\Phi^2 - \Phi\| \leq \eta, \tag{8}$$

where $\|\cdot\|$ on the left is the operator norm of a linear map on $B(\mathcal{H})_{\text{sa}}$ (the order-unit norm $\|x\| = \inf\{t > 0 : -t\mathbf{1} \leq x \leq t\mathbf{1}\}$ being used on $B(\mathcal{H})_{\text{sa}}$ itself). The goal is to replace Φ by an honest idempotent P that is close to Φ and inherits its order properties up to an error of size η . The construction is Kitaev’s; we record the elementary bounds we need with explicit dependence on η .

We work inside the Banach algebra $\text{End}(B(\mathcal{H})_{\text{sa}})$ of bounded linear operators on $B(\mathcal{H})_{\text{sa}}$, whose unit we also denote $\mathbf{1}$. Since Φ is unital positive it is a contraction for the order-unit norm, so $\|\Phi\| = 1$.

Definition 4.1 (Spectral idempotent, (CITED)). Set $S = 2\Phi - \mathbf{1} \in \text{End}(B(\mathcal{H})_{\text{sa}})$. Define

$$R = (S^2)^{-1/2} \tag{9}$$

by the binomial series for $(\mathbf{1} - (\mathbf{1} - S^2))^{-1/2}$, set $\text{sgn}(S) = SR$, and put

$$P = \theta(2\Phi - \mathbf{1}) = \frac{1}{2}(\mathbf{1} + \text{sgn}(S)) = \frac{1}{2}\left(\mathbf{1} + (2\Phi - \mathbf{1})(\mathbf{1} - 4(\Phi - \Phi^2))^{-1/2}\right). \quad (10)$$

This is the spectral idempotent of Φ for the spectral cluster near 1 [8, Eq. (tilde_Φ), Prop. P].

The series defining R converges because $S^2 - \mathbf{1}$ is small: from (8),

$$\|S\| \leq 2\|\Phi\| + 1 \leq 3, \quad \|S^2 - \mathbf{1}\| = 4\|\Phi^2 - \Phi\| \leq 4\eta < 1, \quad (11)$$

the last inequality holding precisely for $\eta < \frac{1}{4}$. Hence R is a well-defined element of $\text{End}(B(\mathcal{H})_{\text{sa}})$ and is a function of S^2 .

Lemma 4.2 (Properties of P , (PROVED)). *There are universal constants $\eta_0 < \frac{1}{4}$ and C such that, whenever $0 \leq \eta \leq \eta_0$, the operator P of Definition 4.1 satisfies:*

- (i) $P^2 = P$ (idempotent);
- (ii) $P(\mathbf{1}) = \mathbf{1}$ (unital);
- (iii) P is a real-linear operator on $B(\mathcal{H})_{\text{sa}}$ (it preserves the self-adjoint part);
- (iv) $\|R - \mathbf{1}\| \leq C\eta$;
- (v) $\delta := \|P - \Phi\| \leq C\eta$;
- (vi) $\|P\| \leq 1 + C\eta$;
- (vii) P is δ -positive: if $a \geq 0$ then $P(a) \geq -\delta\|a\|\mathbf{1}$.

Proof. (i) Because R is a function of S^2 it commutes with S^2 , so $\text{sgn}(S)^2 = S^2 R^2 = S^2 (S^2)^{-1} = \mathbf{1}$. Therefore

$$P^2 = \frac{1}{4}(\mathbf{1} + \text{sgn}(S))^2 = \frac{1}{4}(\mathbf{1} + 2\text{sgn}(S) + \text{sgn}(S)^2) = \frac{1}{2}(\mathbf{1} + \text{sgn}(S)) = P.$$

(ii) Since $\Phi(\mathbf{1}) = \mathbf{1}$ we have $S(\mathbf{1}) = \mathbf{1}$, hence $(S^2 - \mathbf{1})(\mathbf{1}) = 0$. As R is a power series in $S^2 - \mathbf{1}$ with constant term $\mathbf{1}$, this gives $R(\mathbf{1}) = \mathbf{1}$, so $\text{sgn}(S)(\mathbf{1}) = SR(\mathbf{1}) = \mathbf{1}$ and $P(\mathbf{1}) = \frac{1}{2}(\mathbf{1} + \mathbf{1}) = \mathbf{1}$.

(iii) The construction takes place in $\text{End}(B(\mathcal{H})_{\text{sa}})$, and every term in the binomial series is a real-linear endomorphism of $B(\mathcal{H})_{\text{sa}}$. Hence R , $\text{sgn}(S) = SR$, and $P = \frac{1}{2}(\mathbf{1} + \text{sgn}(S))$ all map $B(\mathcal{H})_{\text{sa}}$ to itself.

(iv) Writing $u = \mathbf{1} - S^2$ with $\|u\| \leq 4\eta$ from (11), the binomial series gives $R = \sum_{n \geq 0} \binom{-1/2}{n} (-u)^n$, so

$$\|R - \mathbf{1}\| \leq \sum_{n \geq 1} \left| \binom{-1/2}{n} \right| \|u\|^n \leq \frac{\|u\|/2}{1 - \|u\|} \leq \frac{2\eta}{1 - 4\eta}.$$

Choosing $\eta_0 \leq \frac{1}{8}$ gives $\|R - \mathbf{1}\| \leq 4\eta \leq C\eta$. Here the coefficients satisfy $|\binom{-1/2}{n}| \leq \frac{1}{2}$ for $n \geq 1$, so the tail is dominated by the geometric series $\frac{1}{2} \sum_{n \geq 1} \|u\|^n$.

(v) Since $P - \Phi = \frac{1}{2}(\text{sgn}(S) - S) = \frac{1}{2}S(R - \mathbf{1})$,

$$\delta = \|P - \Phi\| = \frac{1}{2}\|S(R - \mathbf{1})\| \leq \frac{1}{2}\|S\| \|R - \mathbf{1}\| \leq \frac{3}{2}C\eta \leq C\eta,$$

using $\|S\| \leq 3$ and (iv).

(vi) By the triangle inequality and (v), $\|P\| \leq \|\Phi\| + \delta \leq 1 + C\eta$.

(vii) Let $a \geq 0$. Then $\Phi(a) \geq 0$ because Φ is positive, and $\|P(a) - \Phi(a)\| \leq \delta\|a\|$ by (v). For self-adjoint elements, $\|x - y\| \leq \beta$ forces $x \geq y - \beta\mathbf{1}$; applied with $x = P(a)$, $y = \Phi(a) \geq 0$, and $\beta = \delta\|a\|$, this yields $P(a) \geq \Phi(a) - \delta\|a\|\mathbf{1} \geq -\delta\|a\|\mathbf{1}$. $\square$

Remark 4.3 (Kitaev comparison). (CITED; PROVED ADAPTATION) The same operator was introduced by Kitaev for unital completely positive maps, using the completely bounded norm $\|\cdot\|_{\text{cb}}$ in place of the operator norm in (8); the algebraic identities (i)–(ii) and the bounds (iv)–(vi) are norm-independent, so they hold verbatim here [8, Prop. P]. The operator-norm hypothesis used above is what Lemma 4.2 (vii) requires, and it is all that the bridge of the following sections needs.

By Lemma 4.2, P is an idempotent, unital real-linear operator on $B(\mathcal{H})_{\text{sa}}$ with $\delta = \|P - \Phi\| \leq C\eta$. Its range $A = \text{Im } P = \text{Ker}(\mathbf{1} - P)$ is a real subspace of $B(\mathcal{H})_{\text{sa}}$ containing $\mathbf{1}$, and the *projected Jordan product*

$$a \bullet b := P(a \circ b), \quad a, b \in A, \quad (12)$$

will be analysed in Section 5.

Agent B review correction. This section works on the real self-adjoint space $B(\mathcal{H})_{\text{sa}}$. The previous “ $\dagger$ -preserving” formulation was inherited from the complex $B(\mathcal{H})$ statement in Kitaev and was stronger than the domain used here; the bridge proof only needs preservation of $B(\mathcal{H})_{\text{sa}}$.

5 Approximate Jordan algebras

We now introduce the central object of this work: a finite-dimensional real order-unit space carrying a commutative bilinear product that satisfies the JB-algebra axioms of Section 3 only up to a controlled error ε . The structural choice, motivated by the spectral construction of Section 4 (where such an object arises from an almost-idempotent positive map), is that the *order and norm are exact* and only the *product* is approximate. This is the project’s working definition.

Definition 5.1 (ε -JB algebra, order-unit form). (CONSENSUS) Let $\varepsilon \geq 0$. An ε -JB algebra (order-unit form) is a finite-dimensional real order-unit space $(A, \mathbf{1}, \leq)$ with order-unit norm $\|\cdot\|$, together with a commutative bilinear product $\circ: A \times A \rightarrow A$ for which commutativity, $a \circ b = b \circ a$, and the unit law, $\mathbf{1} \circ a = a$, hold *exactly*, and such that for all $a, b \in A$:

$$\|a \circ b\| \leq (1 + \varepsilon) \|a\| \|b\|, \quad (\text{JB1})$$

$$\|a \circ a\| \geq (1 - \varepsilon) \|a\|^2, \quad (\text{JB2})$$

$$a \circ a \geq -\varepsilon \|a\|^2 \mathbf{1}, \quad (\text{JB3})$$

$$\|((a \circ a) \circ b) \circ a - (a \circ a) \circ (b \circ a)\| \leq \varepsilon \|a\|^3 \|b\|. \quad (\text{JB4})$$

We call (JB1) *approximate submultiplicativity*, (JB2) the *lower square norm bound*, (JB3) *approximate positivity of squares*, and (JB4) the *approximate Jordan identity*. The status (CONSENSUS) records that this is the definition agreed between the authors; that it is actually *realised*, with $\varepsilon \leq C\sqrt{\eta}$, by the spectral construction of Section 4 is the separate (PROVED) content of Theorem 6.1.

Remark 5.2. (PROVED FROM DEFINITIONS) At $\varepsilon = 0$ each axiom degenerates to a genuine JB-algebra axiom of Section 3: (JB1) becomes submultiplicativity $\|a \circ b\| \leq \|a\| \|b\|$; (JB2) forces $\|a \circ a\| = \|a\|^2$ (its upper bound is the $\varepsilon = 0$ case of (JB1)), the defining JB norm condition on squares; (JB3) becomes $a \circ a \geq 0$, the formally-real (order) axiom that squares are positive; and (JB4) becomes the exact Jordan identity $(a^2 \circ b) \circ a = a^2 \circ (b \circ a)$ with $a^2 := a \circ a$. Thus an ε -JB algebra is exactly a quantitative relaxation of a JB-algebra in the product alone. The order relation $\leq$, the unit $\mathbf{1}$, the order-unit norm $\|\cdot\|$, and the laws $a \circ b = b \circ a$ and $\mathbf{1} \circ a = a$ are kept *exact*. This is the key structural simplification relative to Kitaev’s ε - C^* algebras [8], where the norm itself is only

approximately multiplicative and the unit may be approximate (e.g. his $\|XY\| \leq (1 + \varepsilon)\|X\|\|Y\|$, $\|X^\dagger X\| \geq (1 - \varepsilon)\|X\|^2$, $|\|I\| - 1| \leq \varepsilon$); here only $\circ$ carries the defect, which removes the approximate-norm bookkeeping at the cost of nothing in the intended application.

Definition 5.3 (δ -Jordan-homomorphism, δ -isomorphism). (CONSENSUS) Let A, B be ε -JB algebras (Definition 5.1) and $\delta \geq 0$. A δ -Jordan-homomorphism is a linear map $v: A \rightarrow B$ that almost preserves the unit and the product:

$$\|v(\mathbf{1}) - \mathbf{1}\| \leq \delta, \quad (13)$$

$$\|v(a \circ b) - v(a) \circ v(b)\| \leq \delta \|a\| \|b\| \quad (a, b \in A). \quad (14)$$

A δ -isomorphism is a bijective δ -Jordan-homomorphism that is moreover an approximate isometry,

$$(1 - \delta) \|a\| \leq \|v(a)\| \leq (1 + \delta) \|a\| \quad (a \in A). \quad (15)$$

This is an algebraic and normed notion of approximate Jordan isomorphism. It does not, by itself, say that v or v^{-1} approximately preserves the positive cone; that order control is an additional requirement in the positive-map factorization problem. The status (CONSENSUS) marks this as the authors' adopted definition — the Jordan (commutative-product) transcription of Kitaev's δ -homomorphism and δ -isomorphism [8].

Agent B review correction. The earlier draft called this equivalently an approximate order-isomorphism. That was not justified by the displayed hypotheses and has been separated from the normed algebraic definition.

6 The bridge theorem

This section contains the central result of the report: from a unital positive almost-idempotent map Φ one constructs, with no use of dilations and no appeal to complete positivity, a space on which the Jordan-algebra axioms hold up to an error $O(\sqrt{\eta})$. The construction and every estimate below are transcribed from the verified internal proof, with notation harmonised to the conventions of this report. The exact (non-perturbative) analogue—that a positive projection onto a unital subspace endows its range with a genuine Jordan structure—is due to Effros and Størmer [3]; the present theorem is the quantitative, almost-idempotent version of that statement.

Throughout, $\mathcal{H}$ is finite-dimensional, $V = B(\mathcal{H})_{\text{sa}}$ carries the order-unit (operator) norm, and the ambient Jordan product is

$$x \circ y = \frac{1}{2}(xy + yx), \quad x, y \in V.$$

We write $a \bullet b := P(a \circ b)$ for the projected Jordan product, where P is the spectral idempotent of Section 4. **All constants C below are universal: they depend on nothing, and in particular are independent of $\dim \mathcal{H}$ (dimension-free).** We are free to decrease the threshold η_0 whenever convenient, in order to absorb an $O(\eta)$ term into an $O(\sqrt{\eta})$ term.

Theorem 6.1 (Algebraic bridge). (PROVED) *There are universal constants $\eta_0 > 0$ and C with the following property. Let $\mathcal{H}$ be finite-dimensional, let $V = B(\mathcal{H})_{\text{sa}}$ with the order-unit/operator norm and Jordan product $x \circ y = \frac{1}{2}(xy + yx)$, and let $\Phi: V \rightarrow V$ be unital and positive with*

$$\|\Phi^2 - \Phi\| \leq \eta \leq \eta_0.$$

Let $P = \theta(2\Phi - \mathbf{1})$ be the spectral idempotent associated to the spectral cluster of Φ near 1 (Section 4), and set

$$A = \text{Im } P, \quad A_+ = A \cap V_+, \quad a \bullet b = P(a \circ b) \quad (a, b \in A).$$

Then $(A, \bullet, \mathbf{1}, A_+)$, with the order structure inherited from V , is an ε -JB order-unit algebra (Section 5) with

$$\varepsilon \leq C\sqrt{\eta}.$$

In particular, for all $a, b \in A$,

$$\mathbf{1} \bullet a = a, \tag{16}$$

$$\|a \bullet b\| \leq (1 + C\eta) \|a\| \|b\|, \tag{17}$$

$$a \bullet a \geq -C\eta \|a\|^2 \mathbf{1}, \tag{18}$$

$$\|a \bullet a\| \geq (1 - C\eta) \|a\|^2, \tag{19}$$

$$\|((a \bullet a) \bullet b) \bullet a - (a \bullet a) \bullet (b \bullet a)\| \leq C\sqrt{\eta} \|a\|^3 \|b\|. \tag{20}$$

Spectral estimate. We first record the basic estimates on P established in Section 4; they are used repeatedly below. Write

$$\delta = \|P - \Phi\|.$$

Since Φ is unital and positive, it is a contraction for the order-unit norm, $\|\Phi\| = 1$. Setting $S = 2\Phi - \mathbf{1}$ in the Banach algebra $\text{End}(V)$ one has

$$\|S\| \leq 3, \quad \|S^2 - \mathbf{1}\| = 4\|\Phi^2 - \Phi\| \leq 4\eta,$$

and $P = \frac{1}{2}(1 + \text{sgn}(S))$ with $\text{sgn}(S) = S(S^2)^{-1/2}$ defined by the convergent binomial series for η_0 small. This P agrees with $\theta(2\Phi - \mathbf{1})$, satisfies $P^2 = P$, fixes the unit, $P(\mathbf{1}) = \mathbf{1}$, and obeys

$$\delta = \|P - \Phi\| \leq C\eta, \quad \|P\| \leq 1 + C\eta. \tag{21}$$

Moreover P is δ -positive:

$$x \geq 0 \implies P(x) \geq -\delta \|x\| \mathbf{1}, \tag{22}$$

because $\Phi(x) \geq 0$ and $\|P(x) - \Phi(x)\| \leq \delta \|x\|$.

Lemma 0: exact order-unit structure on A

Lemma 6.2 (Exact order unit). (PROVED) *The order-unit part of the structure on A is exact. Since $P(\mathbf{1}) = \mathbf{1}$, the range $A = \text{Im } P$ is a real linear subspace of V containing $\mathbf{1}$; equip it with the inherited cone $A_+ = A \cap V_+$. Then $(A, A_+, \mathbf{1})$ is an order-unit space, and its order-unit norm coincides with the ambient operator norm:*

$$\|a\|_{\text{ou}, A} = \inf\{t > 0 : -t\mathbf{1} \leq a \leq t\mathbf{1}\} = \|a\|_{B(\mathcal{H})} \quad (a \in A).$$

Proof. For every $a \in A$ one has $-\|a\| \mathbf{1} \leq a \leq \|a\| \mathbf{1}$ in V , so $\mathbf{1}$ is an order unit for A . The cone is Archimedean because V_+ is closed: if $a + \varepsilon \mathbf{1} \in A_+$ for every $\varepsilon > 0$, then $a \geq 0$ in V , hence $a \in A_+$. The displayed equality of norms is the definition of the order-unit norm together with the fact that, for self-adjoint matrices, the smallest t with $-t\mathbf{1} \leq a \leq t\mathbf{1}$ equals the operator norm. Thus no approximate norm or order structure is introduced; all approximate estimates below concern only the product $\bullet$. $\square$

Standard order estimates

We use only the following order facts, none of which requires complete positivity. Items (O1) and (O2) are the Kadison/Jordan–Schwarz inequalities recalled in Section 2 [7, 11].

- (O1) *Jordan–Schwarz* for unital positive maps on self-adjoint elements: $\Phi(x)^2 \leq \Phi(x^2)$ for $x = x^*$.
- (O2) *Cauchy–Schwarz* for every positive functional ω on $B(\mathcal{H})_{\text{sa}}$: $|\omega(x \circ y)|^2 \leq \omega(x^2) \omega(y^2)$.
- (O3) *Norm monotonicity in the cone*: if $0 \leq x \leq y$ then $\|x\| \leq \|y\|$.
- (O4) *Norm as a supremum over states*: for self-adjoint x , $\|x\| = \sup_{\rho} |\rho(x)|$, the supremum over states ρ on $B(\mathcal{H})$.
- (O5) *Order perturbation*: if x, y are self-adjoint and $\|x - y\| \leq \varepsilon$, then $y - \varepsilon \mathbf{1} \leq x \leq y + \varepsilon \mathbf{1}$.

Lemma 1: the easy axioms

Lemma 6.3 (Easy JB axioms). (PROVED) *For all $a, b \in A$ the identities (16)–(19) hold; that is,*

$$\begin{aligned} \mathbf{1} \bullet a &= a, & a \bullet b &= b \bullet a, & \|a \bullet b\| &\leq (1 + C\eta) \|a\| \|b\|, \\ a \bullet a &\geq -C\eta \|a\|^2 \mathbf{1}, & \|a \bullet a\| &\geq (1 - C\eta) \|a\|^2. \end{aligned}$$

Proof. The unit law $\mathbf{1} \bullet a = P(\mathbf{1} \circ a) = P(a) = a$ and commutativity $a \bullet b = P(a \circ b) = P(b \circ a) = b \bullet a$ are exact, the latter because $\circ$ is commutative. The product bound follows from $\|P\| \leq 1 + C\eta$ in (21) and $\|a \circ b\| \leq \|a\| \|b\|$.

For squares, $a \bullet a = P(a^2)$, and applying (22) to $a^2 \geq 0$ gives $a \bullet a \geq -\delta \|a\|^2 \mathbf{1} \geq -C\eta \|a\|^2 \mathbf{1}$.

For the norm lower bound, since $a = P(a)$ we have $\|\Phi(a) - a\| = \|(\Phi - P)(a)\| \leq \delta \|a\|$. Jordan–Schwarz (O1) gives $\Phi(a^2) \geq \Phi(a)^2$, hence by monotonicity (O3)

$$\|\Phi(a^2)\| \geq \|\Phi(a)^2\| = \|\Phi(a)\|^2 \geq (1 - C\delta) \|a\|^2.$$

Since $\|P(a^2) - \Phi(a^2)\| \leq \delta \|a\|^2$, we conclude

$$\|a \bullet a\| = \|P(a^2)\| \geq (1 - C\eta) \|a\|^2.$$

Thus only the Jordan identity (20) remains to be proved. □

Lemma 2: the first insertion estimate

For every state ρ on $B(\mathcal{H})$ set

$$\omega = \rho \circ \Phi, \quad \|x\|_{\omega}^2 = \omega(x^2). \quad (23)$$

Since Φ is positive, ω is a positive functional and $\|\cdot\|_{\omega}$ is a seminorm on V .

Lemma 6.4 (First insertion estimate). (PROVED) (i) (Almost-contraction.) *For all $x \in V$,*

$$\|Px\|_{\omega}^2 \leq \|x\|_{\omega}^2 + C\eta \|x\|^2. \quad (24)$$

(ii) (Almost-orthogonality.) *For $u \in \text{Im } P$ and $n \in \text{Ker } P$,*

$$|\omega(u \circ n)| \leq C\sqrt{\eta} \|u\| \|n\|. \quad (25)$$

(iii) (First insertion (FI).) *For all $x \in V$ and $b \in A$,*

$$\|P(Px \circ b) - P(x \circ b)\| \leq C\sqrt{\eta} \|x\| \|b\| \quad (\text{FI}). \quad (26)$$

Proof. (i) We chain three inequalities:

$$\rho \Phi((Px)^2) \leq \rho \Phi((\Phi x)^2) + C\eta \|x\|^2 \leq \rho \Phi^2(x^2) + C\eta \|x\|^2 \leq \rho \Phi(x^2) + C\eta \|x\|^2.$$

The first inequality uses $\|Px - \Phi x\| \leq \delta \|x\|$ together with the elementary bound $\|y^2 - z^2\| \leq 2 \max(\|y\|, \|z\|) \|y - z\|$ and $\|P\|, \|\Phi\| \leq C$: applying the state $\rho \circ \Phi$ to the self-adjoint difference of the two squares costs at most its operator norm, which is $O(\eta) \|x\|^2$. The middle inequality is Jordan–Schwarz (O1) for Φ applied to Φx ; the last uses $\|\Phi^2 - \Phi\| \leq \eta$. This is (24).

(ii) Since P is an exact idempotent, (24) forces almost-orthogonality of $\text{Im } P$ and $\text{Ker } P$ in every seminorm $\|\cdot\|_\omega$. Put $c = \omega(u \circ n)$ and $a = \|n\|_\omega^2$. Because $P(u + tn) = u$ for every real t , (24) applied to $x = u + tn$ gives

$$\|u\|_\omega^2 \leq \|u + tn\|_\omega^2 + K\eta \|u + tn\|^2 = \|u\|_\omega^2 + 2tc + t^2a + K\eta \|u + tn\|^2,$$

where K is the constant of (24). Choose the sign of t opposite to the sign of c and write $s = |t|$; using $\|u + tn\|^2 \leq 2\|u\|^2 + 2s^2\|n\|^2$ we obtain, for all $s > 0$,

$$2s|c| \leq s^2(a + 2K\eta \|n\|^2) + 2K\eta \|u\|^2.$$

Optimising over $s > 0$ yields

$$|c| \leq C\sqrt{\eta} \|u\| \sqrt{a + \eta \|n\|^2} \leq C\sqrt{\eta} \|u\| \|n\|,$$

the last step using $a = \|n\|_\omega^2 \leq \|n\|^2$. This is (25).

(iii) Let $x \in V$, $b \in A$, and set $n = x - Px \in \text{Ker } P$, so $\|n\| \leq C\|x\|$. For every state ρ , with $\omega = \rho \circ \Phi$,

$$|\rho P(n \circ b)| \leq |\rho \Phi(n \circ b)| + C\eta \|n\| \|b\| = |\omega(n \circ b)| + C\eta \|n\| \|b\| \leq C\sqrt{\eta} \|x\| \|b\|,$$

where the first step replaces P by Φ at cost $\delta \|n \circ b\| \leq C\eta \|n\| \|b\|$ and the last applies (25) with $u = b \in \text{Im } P$. Since $P(n \circ b)$ is self-adjoint and $P(x \circ b) - P(Px \circ b) = P(n \circ b)$, taking the supremum over states (O4) gives (26). $\square$

Lemma 3: square holes are approximately null

For $r \in A$ define the *square hole*

$$q_r = P(r^2) - r^2 \in \text{Ker } P,$$

the inclusion holding because $P(q_r) = P^2(r^2) - P(r^2) = 0$.

Lemma 6.5 (Square holes are almost null). (PROVED) *For every $r \in A$ and every state seminorm $\omega = \rho \circ \Phi$,*

$$q_r \geq -C\eta \|r\|^2 \mathbf{1}, \tag{27}$$

$$\|P(q_r^2)\| \leq C\eta \|r\|^4, \tag{28}$$

$$\|q_r\|_\omega \leq C\sqrt{\eta} \|r\|^2. \tag{29}$$

Proof. Since $\|\Phi(r) - r\| \leq \delta \|r\|$, Jordan–Schwarz (O1) and order perturbation (O5) give

$$\Phi(r^2) \geq \Phi(r)^2 \geq r^2 - C\delta \|r\|^2 \mathbf{1},$$

the second inequality from $\|\Phi(r)^2 - r^2\| \leq C\delta \|r\|^2$. Replacing $\Phi(r^2)$ by $P(r^2)$ costs another $\delta \|r\|^2$ in the same order sense, which yields $P(r^2) \geq r^2 - C\delta \|r\|^2 \mathbf{1}$, i.e. (27) (recall $\delta \leq C\eta$).

Choose $\gamma = C\delta \|r\|^2$ so that $y = q_r + \gamma \mathbf{1} \geq 0$. Also $\|q_r\| \leq C\|r\|^2$ and hence $\|y\| \leq C\|r\|^2$. Since $P(q_r) = 0$,

$$\|\Phi(q_r)\| = \|(\Phi - P)(q_r)\| \leq C\delta \|r\|^2,$$

so $0 \leq \Phi(y)$ with $\|\Phi(y)\| \leq C\delta \|r\|^2$. Because $0 \leq y \leq C\|r\|^2 \mathbf{1}$, functional calculus gives $0 \leq y^2 \leq C\|r\|^2 y$, and positivity of Φ together with monotonicity (O3) yields

$$\|\Phi(y^2)\| \leq C\|r\|^2 \|\Phi(y)\| \leq C\delta \|r\|^4.$$

Finally $q_r^2 \leq 2y^2 + 2\gamma^2 \mathbf{1}$, so $\|\Phi(q_r^2)\| \leq C\delta \|r\|^4$; the term $\gamma^2 \mathbf{1}$ is $O(\delta^2 \|r\|^4)$ and is absorbed into $O(\delta \|r\|^4)$ after decreasing η_0 . Replacing Φ by P costs at most $\delta \|q_r^2\| \leq C\delta \|r\|^4$, proving (28). Estimate (29) follows since $\|q_r\|_\omega^2 = \rho \Phi(q_r^2) \leq \|\Phi(q_r^2)\| \leq C\eta \|r\|^4$. $\square$

Lemma 4: polarised hole estimates

For $r, s \in A$ set the *product hole*

$$q_{r,s} = P(r \circ s) - r \circ s = -h_{r,s},$$

a symmetric bilinear function of (r, s) lying in $\text{Ker } P$.

Lemma 6.6 (Polarised hole estimates). (PROVED) *For all $r, s, u, v \in A$, all $z \in V$, and every state seminorm $\omega = \rho \circ \Phi$,*

$$\|q_{r,s}\|_\omega \leq C\sqrt{\eta} \|r\| \|s\|, \quad (30)$$

$$\|q_{r,s}\| \leq C\|r\| \|s\|, \quad (31)$$

$$\|P(q_{r,s} \circ q_{u,v})\| \leq C\eta \|r\| \|s\| \|u\| \|v\| \quad (\text{HH}), \quad (32)$$

$$\|P(q_{r,s} \circ z)\| \leq C\sqrt{\eta} \|r\| \|s\| \|z\| \quad (\text{HZ}). \quad (33)$$

Proof. Estimate (30). For $\lambda > 0$, polarisation of the symmetric bilinear map $(r, s) \mapsto q_{r,s}$ gives

$$\lambda q_{r,s} = q_{\lambda r, s} = \frac{1}{4}(q_{\lambda r+s} - q_{\lambda r-s}),$$

where $q_x = P(x^2) - x^2$ is the square hole of Lemma 6.5. Hence (29) gives

$$\|q_{r,s}\|_\omega \leq C\sqrt{\eta}(\lambda \|r\|^2 + \lambda^{-1} \|s\|^2).$$

If $r = 0$ or $s = 0$ this is trivial; otherwise put $\lambda = \|s\|/\|r\|$ to get (30). The crude bound (31) is immediate from $\|P\| \leq C$ and $\|r \circ s\| \leq \|r\| \|s\|$.

Estimate (32). For every state ρ , Cauchy–Schwarz (O2) for the positive functional $\omega = \rho \circ \Phi$ gives

$$|\rho \Phi(q_{r,s} \circ q_{u,v})| \leq \|q_{r,s}\|_\omega \|q_{u,v}\|_\omega \leq C\eta \|r\| \|s\| \|u\| \|v\|$$

by (30). Replacing Φ by P costs at most $\delta \|q_{r,s}\| \|q_{u,v}\|$, which has the same bound after increasing C (using (31)). Since $P(q_{r,s} \circ q_{u,v})$ is self-adjoint, the state supremum (O4) is its operator norm.

Estimate (33). For $z \in V$ we have $\|z\|_\omega \leq \|z\|$, because $z^2 \leq \|z\|^2 \mathbf{1}$ and Φ is positive unital. Hence, as above, for every state ρ ,

$$|\rho \Phi(q_{r,s} \circ z)| \leq \|q_{r,s}\|_\omega \|z\|_\omega \leq C\sqrt{\eta} \|r\| \|s\| \|z\|.$$

Replacing Φ by P costs at most $\delta \|q_{r,s}\| \|z\| \leq C\eta \|r\| \|s\| \|z\|$, which is absorbed into the displayed $O(\sqrt{\eta})$ bound. Taking the state supremum (O4) gives (33). $\square$

Lemma 5: one-hole contexts

Lemma 6.7 (One-hole contexts). (PROVED) *Let $h = h_{r,s} = r \circ s - P(r \circ s)$ be a range-product hole, $r, s \in A$, and let $t, u \in A$. Then*

$$\|P((h \circ t) \circ u)\| \leq C\sqrt{\eta} \|r\| \|s\| \|t\| \|u\| \quad (5.1),$$

$$\|P(h \circ (t \circ u))\| \leq C\sqrt{\eta} \|r\| \|s\| \|t\| \|u\| \quad (5.2).$$

Proof. Estimate (34). Put $y = h \circ t$. Since $h = -q_{r,s}$, (33) (with $z = t$) gives $\|P(y)\| \leq C\sqrt{\eta} \|r\| \|s\| \|t\|$, hence

$$\|P(P(y) \circ u)\| \leq \|P\| \|P(y)\| \|u\| \leq C\sqrt{\eta} \|r\| \|s\| \|t\| \|u\|.$$

For the kernel part $y - P(y)$, apply (26) with $x = y$ and $b = u$, using the crude bound $\|y\| \leq C\|r\| \|s\| \|t\|$:

$$P((y - P(y)) \circ u) = P(y \circ u) - P(P(y) \circ u)$$

has the same $O(\sqrt{\eta} \|r\| \|s\| \|t\| \|u\|)$ bound. Adding the two parts gives (34).

Estimate (35). Write $t \circ u = P(t \circ u) + h_{t,u}$. The first term gives $P(h \circ P(t \circ u))$, controlled by (33) because $P(t \circ u) \in A$ and $h = -q_{r,s}$; the second gives $P(h \circ h_{t,u}) = P(q_{r,s} \circ q_{t,u})$, controlled by (32), hence $O(\eta \|r\| \|s\| \|t\| \|u\|)$ and absorbed into the $O(\sqrt{\eta})$ target. $\square$

Assembly: the Jordan identity

We now prove (20), the last remaining axiom. The strategy is to push both sides through the exact Jordan identity of the ambient product $\circ$ recalled in Section 3, paying only $O(\sqrt{\eta})$ for each hole created in the process.

Proposition 6.8 (Approximate Jordan identity). (PROVED) *For all $a, b \in A$,*

$$\|((a \bullet a) \bullet b) \bullet a - (a \bullet a) \bullet (b \bullet a)\| \leq C\sqrt{\eta} \|a\|^3 \|b\|.$$

Proof. Fix $a, b \in A$ and write

$$p = P(a^2) = a^2 - h_{a,a}, \quad q = P(b \circ a) = b \circ a - h_{b,a},$$

so that $p, q \in A$ and, by the crude bounds,

$$\|p\| \leq C\|a\|^2, \quad \|q\| \leq C\|a\| \|b\|.$$

The left side. Let $l = h_{p,b} = p \circ b - P(p \circ b)$. Then

$$((a \bullet a) \bullet b) \bullet a = P(P(p \circ b) \circ a) = P((p \circ b) \circ a) - P(l \circ a).$$

Here $l \in \text{Ker } P$, since $P(l) = P(p \circ b) - P^2(p \circ b) = 0$, and

$$\|l\| \leq \|p \circ b\| + \|P(p \circ b)\| \leq C\|a\|^2 \|b\|.$$

As $P(l) = 0$, (26) applied with $x = l$ and range element a gives

$$\|P(l \circ a)\| \leq C\sqrt{\eta} \|a\|^3 \|b\| \quad (6.1).$$

Next, using $p = a^2 - h_{a,a}$,

$$P((p \circ b) \circ a) = P((a^2 \circ b) \circ a) - P((h_{a,a} \circ b) \circ a),$$

and the last term is $O(\sqrt{\eta} \|a\|^3 \|b\|)$ by (34) with $h = h_{a,a}$, $t = b$, $u = a$. Combining with (36),

$$((a \bullet a) \bullet b) \bullet a = P((a^2 \circ b) \circ a) + O(\sqrt{\eta} \|a\|^3 \|b\|) \quad (\text{L}). \quad (37)$$

The right side. Expanding both holes,

$$(a \bullet a) \bullet (b \bullet a) = P(p \circ q) = P((a^2 - h_{a,a}) \circ (b \circ a - h_{b,a})),$$

which equals

$$P(a^2 \circ (b \circ a)) - P(h_{a,a} \circ (b \circ a)) - P(a^2 \circ h_{b,a}) + P(h_{a,a} \circ h_{b,a}).$$

We bound the three error terms.

- $\|P(h_{a,a} \circ (b \circ a))\| \leq C\sqrt{\eta} \|a\|^3 \|b\|$ by (35) with $h = h_{a,a}$, $t = b$, $u = a$.
- For the term $P(a^2 \circ h_{b,a})$, decompose $a^2 = p + h_{a,a}$:

$$P(a^2 \circ h_{b,a}) = P(p \circ h_{b,a}) + P(h_{a,a} \circ h_{b,a}).$$

Since $h_{b,a} \in \text{Ker } P$ and $p \in A$, (26) applied with $x = h_{b,a}$ and range element p gives $P(P(h_{b,a}) \circ p) = 0$, whence

$$\|P(p \circ h_{b,a})\| \leq C\sqrt{\eta} \|p\| \|h_{b,a}\| \leq C\sqrt{\eta} \|a\|^3 \|b\|.$$

- Both occurrences of $P(h_{a,a} \circ h_{b,a}) = P(q_{a,a} \circ q_{b,a})$ are bounded by (32): $\|P(h_{a,a} \circ h_{b,a})\| \leq C\eta \|a\|^3 \|b\|$, which for $\eta \leq \eta_0$ is absorbed into the $O(\sqrt{\eta})$ error.

Therefore

$$(a \bullet a) \bullet (b \bullet a) = P(a^2 \circ (b \circ a)) + O(\sqrt{\eta} \|a\|^3 \|b\|) \quad (\text{R}). \quad (38)$$

Cancellation of the leading term. The ambient product $\circ$ satisfies the *exact* Jordan identity (Section 3):

$$(a^2 \circ b) \circ a = a^2 \circ (b \circ a).$$

Hence the leading terms $P((a^2 \circ b) \circ a)$ in (37) and $P(a^2 \circ (b \circ a))$ in (38) are identical. Subtracting (38) from (37) leaves only the error terms:

$$\|((a \bullet a) \bullet b) \bullet a - (a \bullet a) \bullet (b \bullet a)\| \leq C\sqrt{\eta} \|a\|^3 \|b\|. \quad \square$$

Proof of Theorem 6.1. Lemma 6.2 establishes that $(A, A_+, \mathbf{1})$ is an order-unit space with order-unit norm equal to the ambient operator norm, so the order structure is exact. Lemma 6.3 gives the unit law, commutativity, the product bound (17), the square positivity (18), and the square-norm lower bound (19). Proposition 6.8 gives the approximate Jordan identity (20). Together these are exactly the axioms of an ε -JB order-unit algebra (Section 5) with $\varepsilon \leq C\sqrt{\eta}$. All constants are universal and dimension-free. $\square$

Remark 6.9. (CITED) The non-perturbative limit $\eta = 0$ recovers the exact statement of Effros and Størmer [3]: a positive unital projection P on a unital JC-algebra makes its range a JC-algebra under $a \bullet b = P(a \circ b)$. Theorem 6.1 is the quantitative, almost-idempotent version, in which every hole estimate degrades gracefully as $O(\sqrt{\eta})$ or $O(\eta)$ and the exact Jordan identity of the ambient algebra supplies the leading-order cancellation.

7 The structure-theory programme

The preceding sections produced, from an almost-idempotent positive map, an ε -JB algebra: a finite-dimensional real space with a commutative product $a \bullet b := P(a \circ b)$, a unit $\mathbf{1}$, and an order-unit norm, satisfying the JB axioms up to an error of order ε (see Section 5). The goal of the structure-theory programme is to remove the error entirely: to show that such an object is, up to a controlled deformation, a genuine JB-algebra. This section states that goal honestly as an open target, records the proof strategy, and isolates the one quantitative obstruction together with candidate routes around it. Nothing below is claimed as a completed proof; every named target carries the tag (OPEN).

7.1 The target

The model to imitate is Kitaev’s structure theorem for approximate C^* -algebras. In his terminology an ε - C^* algebra is a finite-dimensional space with a product, involution and norm satisfying the C^* axioms up to ε , and a δ -isomorphism is a linear bijection preserving unit and product up to a relative error δ . His main theorem reads, in source notation after harmless macro expansion:

“Any finite-dimensional ε - C^* algebra $\mathcal{A}$ is $O(\varepsilon)$ -isomorphic to some C^* algebra $\mathcal{B}$. (Importantly, the implicit constant in $O(\varepsilon)$ does not depend on $\mathcal{A}$ or its dimensionality.)”

[8], Theorem `th_main`. The decisive feature is the last sentence: the constant is *dimension-free*. The Jordan analogue we are after is the following.

Open Problem 7.1 (Jordan structure theorem). (OPEN) *There are universal constants $C, \varepsilon_0 > 0$ (independent of dimension) such that for every $\varepsilon < \varepsilon_0$ and every finite-dimensional ε -JB algebra $\mathcal{A}$ there exist a genuine finite-dimensional JB-algebra $\mathcal{B}$ and a linear bijection $v: \mathcal{B} \rightarrow \mathcal{A}$ that is a $C\varepsilon$ -Jordan-isomorphism, i.e.*

$$\|v(\mathbf{1}_{\mathcal{B}}) - \mathbf{1}_{\mathcal{A}}\| \leq C\varepsilon, \quad \|v(b \bullet b') - v(b) \bullet v(b')\| \leq C\varepsilon \|b\| \|b'\|,$$

and $(1 - C\varepsilon)\|b\| \leq \|v(b)\| \leq (1 + C\varepsilon)\|b\|$ for all $b, b' \in \mathcal{B}$.

This is the analogue of Kitaev’s statement with the associative product replaced by the Jordan product $\bullet$ and “ C^* algebra” replaced by “JB-algebra”. As of the present report it is a target, not a theorem. What we *can* state is the strategy and the precise point at which the dimension-free constant is in question.

7.2 The strategy

Kitaev’s proof builds the model algebra $\mathcal{B}$ incrementally and controls the error at each step by a cohomological argument. The Jordan programme mirrors the four moving parts.

Vocabulary for the programme. An *idempotent element* is an element e with $e \bullet e = e$. A *Jordan frame* is a finite list of nonzero idempotents that are pairwise orthogonal, $e_i \bullet e_j = 0$ for $i \neq j$, and sum to the unit; it plays the role of diagonal matrix units. The *Peirce decomposition* is the splitting of the algebra into the eigenspaces of multiplication by an idempotent. A *cochain* is a multilinear correction term; a *cocycle* is a defect that satisfies the compatibility identity forced by the Jordan identity; a *coboundary* is a defect removable by changing the comparison map. A *contracting homotopy* or *splitting* is a bounded linear recipe that turns a cocycle into such a correction. The

projective tensor norm records the cost of writing a tensor as a sum of simple tensors; dimension-free control means this cost is bounded by a universal constant.

(a) Approximate idempotents and a Jordan frame. A genuine JB-algebra is coordinatized by a complete system of orthogonal idempotents (a *Jordan frame*). The first step produces such idempotents *approximately*: elements $e = e^*$ with $\|e \bullet e - e\|$ small and not too close to 0 or **1**. In the associative case these are the nontrivial projections produced by a fixed-point argument [8]; the spectral idempotent P of Section 4 is the map-level device that produced the ε -JB product. It does *not* by itself produce idempotent elements in A ; constructing approximate Jordan-frame elements is a separate open step. The Jordan frame would then be refined by repeatedly splitting an approximate idempotent into two, exactly as the associative model algebra is grown from $\mathbb{C}$ to $\mathbb{C} \oplus \mathbb{C}$ and onward.

Agent B review correction. The earlier draft blurred the idempotent linear map P with idempotent elements of the approximate algebra. Layer 1 still needs its own approximate-frame construction.

(b) Approximate Peirce decomposition. A genuine idempotent e in a Jordan algebra induces the Peirce decomposition $\mathcal{A} = \mathcal{A}_1 \oplus \mathcal{A}_{1/2} \oplus \mathcal{A}_0$ into the 1, $\frac{1}{2}$, 0 eigenspaces of the multiplication operator $L_e: a \mapsto e \bullet a$. For an approximate idempotent the operator L_e has spectrum clustered near $\{0, \frac{1}{2}, 1\}$, and the associated spectral projections give an *approximate* Peirce decomposition with $O(\varepsilon)$ cross-terms. This decomposes $\mathcal{A}$ into pieces on which the product is closer to the model multiplication rules.

(c) Approximate coordinatization. On the joint Peirce spaces of a frame of rank ≥ 3 , the Jordan coordinatization theorem (the Jordan–von Neumann–Wigner coordinatization of $H_n(\mathfrak{C})$, [6]) identifies the algebra with hermitian matrices over a coordinate $*$ -algebra. The approximate version reads off coordinate elements from the approximate Peirce pieces and checks the coordinate relations up to $O(\varepsilon)$. This step delivers a candidate genuine JB-algebra $\mathcal{B}$ together with a linear inclusion $v: \mathcal{B} \rightarrow \mathcal{A}$ that is multiplicative up to a (possibly large) error δ .

(d) Error reduction. The inclusion v from (c) is only a δ -inclusion with $\delta \gg \varepsilon$. The final step replaces v by a nearby map whose defect is $O(\varepsilon)$, *uniformly in the dimension*. This is the cohomological heart of the argument and is treated separately below.

The algebraic inputs to (d) are classical and we cite them as facts. For a finite-dimensional Jordan algebra J of characteristic 0 and a J -module M , the two Whitehead lemmas assert the vanishing of the first two cohomology groups in the semisimple/separable case.

Theorem 7.2 (Whitehead lemmas for Jordan algebras, (SOURCE-EXTRACTED; PENDING PRIMARY BYTE-CHECK)). *Let J be a finite-dimensional semisimple Jordan algebra over a field of characteristic 0 and let M be a finite-dimensional J -module.*

1. (First Whitehead lemma.) *Every derivation $f: J \rightarrow M$ is inner; equivalently $H^1(J, M) = 0$.*
2. (Second Whitehead lemma.) *If moreover J is separable, every Jordan 2-cocycle is a coboundary; equivalently $H^2(J, M) = 0$.*

The first lemma is due to Jacobson; the second to Penico, who used it to prove the Wedderburn principal theorem for Jordan algebras.

The first part is the Jordan analogue of the first Whitehead lemma, recorded as Theorem 1.1 of [2]: “Every linear $f: J \rightarrow M$ with $f(ab) = f(a)b + af(b)$ is inner.” The second part is the Jordan analogue of the second Whitehead lemma, Theorem 1.2 of [2], attributed there to [9]: “Every Jordan

2-cocycle f is a coboundary.” The vanishing of H^1 and H^2 is what would make the exact correction step possible *in principle*; the issue is the size of the correction.

The vanishing has a concrete mechanism, which is the second cited input. The *multiplication algebra* $M(J) \subseteq \text{End}(J)$ generated by $\mathbf{1}$ and the operators $L_a: x \mapsto a \bullet x$ is, for semisimple J , a semisimple *associative algebra*; hence every J -module is completely reducible and the cohomology splits. This is the engine behind Theorem 7.2.

Proposition 7.3 (Compactness of the automorphism group, (SOURCE-EXTRACTED; PENDING PRIMARY BYTE-CHECK)). *For a finite-dimensional Euclidean (formally real) Jordan algebra J the automorphism group $\text{Aut}(J)$ is compact: it is a closed subgroup of the orthogonal group of the trace inner product, and every automorphism is an isometry of that inner product. Consequently normalized Haar averaging over $\text{Aut}(J)$ is available, producing invariant inner products and invariant tensors.*

Proposition 7.3 is the analytic input recorded in [4]: Jordan automorphisms of a Euclidean Jordan algebra preserve the trace form, so $\text{Aut}(J)$ is a closed, hence compact, subgroup of an orthogonal group. This is the Jordan counterpart of the compact unitary group that Kitaev averages over.

Agent B review provenance flag. Theorem 7.2 and proposition 7.3 are registered in the provenance ledger through the local ingestion file `agent-A/notes/ingestion-results-2026-06-01.json`. The Chu–Russo primary source is present locally as a PDF, but the Whitehead quotes have not yet been independently byte-matched against extracted primary text; Faraut–Koranyi compactness is likewise extraction-level provenance in this draft.

7.3 The dimension-free obstruction and candidate routes

We now isolate the single quantitative difficulty in step (d), state the obstruction precisely, and describe the resolution proposed in the project’s internal notes. This subsection is **original** to the project and is explicitly a **programme**, marked (OPEN); it is being tested numerically.

The cochain setup. Write $g(b, b') := v(b \bullet b') - v(b) \bullet v(b')$ for the multiplicativity defect of the inclusion $v: \mathcal{B} \rightarrow \mathcal{A}$ from step (c); it is symmetric with $\|g\| \leq \delta$. Linearizing the Jordan identity (which $\mathcal{B}$ satisfies exactly and $\mathcal{A}$ only up to ε) shows that g is an ε -approximate Jordan 2-cocycle: the cocycle identity holds up to $O(\varepsilon) \cdot \|g\|^3$. By Theorem 7.2 (with $J = \mathcal{B}$) an exact cocycle would be an exact coboundary, $g = d^1 h$ for a linear $h: \mathcal{B} \rightarrow \mathcal{A}$, where $(d^1 h)(b, b') = v(b) \bullet h(b') + h(b) \bullet v(b') - h(b \bullet b')$. Setting $\tilde{v} := v - h$ replaces the defect by $\tilde{g} = g - d^1 h + O(\delta^2) = O(\delta^2 + K\varepsilon)$, where $K = \|s\|$ is the norm of the splitting operator s that produces $h = sg$. Iterating the resulting Newton step $\delta \mapsto O(\delta^2 + K\varepsilon)$ drives any sufficiently small δ down to $O(K\varepsilon)$. The theorem therefore reduces to one quantitative question: *is K dimension-free?*

The obstruction. The splitting s is built from a *diagonal* (Casimir) tensor $D = \sum_j u_j \otimes u'_j$ of the multiplication algebra $M(\mathcal{B})$; the natural candidate is $D = \sum_i e_i \otimes e^i$ formed from a pair of trace-form-dual bases. The relevant norm is the projective tensor norm $\|\cdot\|_\Lambda$, an infimum over representations. Evaluated on the *basis representation*, this naive Casimir has

$$\|D\|_\Lambda \sim \dim \mathcal{B},$$

so the resulting K grows with the dimension and the Newton iteration only converges when $\varepsilon < c/\dim \mathcal{B}$. This is exactly the dimension-dependent bound Kitaev encounters for the naive associative diagonal: he notes that “naive constructions of the Haar measure (or just the diagonal) in the ε -associative setting have error bounds proportional to $n = \dim \mathcal{A}$ ” [8].

Candidate averaging mechanism (original). The project note registered in `PROVENANCE.md` as **A-ER** proposes that $\|\cdot\|_\Lambda$ should be estimated using a compact-group representation rather than a basis representation. In the associative case Kitaev’s diagonal is the group average

$$D = \int_{U(n)} U^\dagger \otimes U \, dU = \frac{1}{n} \text{SWAP},$$

[8]. The basis representation of $\frac{1}{n} \text{SWAP}$ gives $\|D\|_\Lambda \leq \frac{1}{n} \cdot n^2 = n$, but the *averaging* representation gives

$$\|D\|_\Lambda \leq \int \|U^\dagger\|_{\text{op}} \|U\|_{\text{op}} \, dU = \int 1 \, dU = 1.$$

It is the *same* element; only the compact-group average exhibits the $O(1)$ projective norm. The proposed Jordan analogue tries to imitate this by using, not $\mathcal{B} \otimes \mathcal{B}$ directly, but the *separability idempotent* $e = \sum_i u_i \otimes u'_i \in M(\mathcal{B}) \otimes M(\mathcal{B})^{\text{op}}$ of the semisimple associative algebra $M(\mathcal{B})$. In the trace/Frobenius model one has a compact unitary group $U(M(\mathcal{B}))$ and a dimension-free-looking average of the form

$$e = \int_{U(M(\mathcal{B}))} u^\dagger \otimes u \, du, \quad \|e\|_\Lambda \leq \int \|u^\dagger\| \|u\| \, du = 1,$$

which explains the favourable Frobenius-norm numerics below. This is not yet a proof in the order-unit/operator norm required by Open Problem 7.1: the unitary group of $M(\mathcal{B})$ preserves the trace norm, while the order-unit norm is controlled by the generally smaller Jordan automorphism group. The open task is to turn this candidate average, or an incremental replacement for it, into a contracting homotopy s with order-unit-norm bound $\|s\| = K = O(1)$.

Open Problem 7.4 (Dimension-free error reduction). (OPEN; PROGRAMME TARGET). *If the compact-group-averaged separability idempotent of the multiplication algebra $M(\mathcal{B})$ can be converted into a contracting homotopy s for Jordan 2-cocycles with order-unit-norm bound $\|s\| = K = O(1)$, independent of $\dim \mathcal{B}$, then the Newton iteration $\delta \mapsto O(\delta^2 + K\varepsilon)$ of step (d) terminates at $\delta = O(\varepsilon)$ with a dimension-free constant, which is what Open Problem 7.1 requires.*

The status of Open Problem 7.4 is the following. The two algebraic inputs — vanishing cohomology (Theorem 7.2) and a compact averaging group (Proposition 7.3) — are cited facts. The mechanism that makes the constant dimension-free is established for the associative diagonal by Kitaev; its transfer to the Jordan setting has been examined numerically, with a result that is informative and must be reported precisely.

Numerical evidence (COMPUTATIONAL EVIDENCE; NOT THEOREM), **and the precise remaining gap.** We computed the smallest nonzero singular value of d^1 , hence $\|s\| = 1/\sigma_{\min}(d^1)$, for $J = H_n(\mathbb{C})$, $J = H_n(\mathbb{R})$, and the spin factors V_n , in two norms on the cochain spaces. (The computation is validated by $\dim \text{Ker } d^1 = \dim \text{Der}(J)$: equal to $n^2 - 1$ for $H_n(\mathbb{C})$ and $n(n-1)/2$ for V_n , both matched exactly.) The outcome splits along the choice of norm:

- *Trace (Frobenius) inner-product norm:* $\|s\|$ is **bounded — in fact ≤ 1 and decreasing** in every family ($H_n(\mathbb{C})$: 0.91, 0.78, 0.69, 0.62, 0.57 for $n = 2, \dots, 6$; V_n : convergent to ≈ 0.62). In this norm the dimension-free splitting exists.
- *Order-unit (operator) norm — the norm Open Problem 7.1 actually requires:* certified lower bounds on $\|s\|_{\text{op}}$ for $H_n(\mathbb{C})$ grow only *slowly* ($1.4 \rightarrow 1.8$ for $n = 2, \dots, 6$, with $\|s\|_{\text{op}}/\sqrt{n}$ and $\|s\|_{\text{op}}/n$ both decreasing), while a crude upper bound grows like n . **Boundedness in the operator norm is therefore unresolved** at this range.

The discrepancy is structural and locates the gap exactly. The Frobenius bound comes from averaging over the unitary group of the multiplication algebra $M(J)$, whose elements are Frobenius isometries. The order-unit norm, however, is preserved only by the Jordan automorphism group $\text{Aut}(J)$, which is *strictly smaller* than $U(M(J))$. The $O(1)$ projective-norm representation of the separability idempotent uses the larger group, so it does not by itself give an operator-norm bound. Closing the gap requires one of: (i) an $\text{Aut}(J)$ -only construction of the splitting carrying an operator-norm bound; (ii) Kitaev’s incremental route, applying the splitting only to a controlled algebra $\mathcal{B}$ built one step at a time, so that the dimension never enters; or (iii) a formulation in the trace norm with an explicit, possibly dimension-dependent, conversion. Independently of the norm, one must still write the explicit contraction $h = sg$, verify $d^1 h = g + O(\varepsilon)$ on ε -approximate cocycles, and absorb the approximate-module errors (the module $\mathcal{A}$ is itself only an ε -JB algebra). Until these are complete, Open Problem 7.1 and Open Problem 7.4 remain (OPEN). The numerical detail is recorded in `agent-A/experiments/jordan-coboundary/REPORT.md`.

8 Status and open problems

This section records what is proved, what is cited, and what is open. We separate the two layers of the programme: the *bridge* (Section 6), which turns an almost-idempotent positive map into an approximate Jordan algebra, and the *structure theory* (Section 7), which would turn an approximate Jordan algebra back into a genuine one. Throughout, $\Phi: B(\mathcal{H})_{\text{sa}} \rightarrow B(\mathcal{H})_{\text{sa}}$ is unital and positive ($*$ -preserving) with $\|\Phi^2 - \Phi\| \leq \eta$, $P = \theta(2\Phi - \mathbf{1})$ the spectral idempotent close to Φ , $A = \text{Im } P$, and $a \bullet b := P(a \circ b)$.

8.1 The exponent: $\sqrt{\eta}$ versus η

The bridge of Section 6 produces an ε -JB-algebra with $\varepsilon = O(\sqrt{\eta})$, and this exponent is unconditional. The Kitaev-strength exponent $\varepsilon = O(\eta)$ is available in the UCP/cb setting treated by Kitaev, and in any positive-map setting that supplies a compatible lifted-UCP dilation model. For arbitrary positive maps, and for decomposable maps without such compatible data, it remains conjectural. The following remark explains why.

Remark 8.1 (Why CP gives η and positivity gives only $\sqrt{\eta}$; (PROVED FOR THE BRIDGE AT $\sqrt{\eta}$; η OPEN)). Both regimes are governed by a single “hole” quantity: the failure of a single inserted Φ (or P) to be invisible. For a unital positive Φ a single insertion is controlled only in the seminorm of the state $\omega = \rho \circ \Phi$, where Jordan–Schwarz gives a defect of squared size $O(\eta)$, hence norm $O(\sqrt{\eta})$. This is the source of the unconditional $\varepsilon = O(\sqrt{\eta})$. Kitaev’s $O(\eta)$ does *not* arise from a single insertion: the associativity defect is an intrinsically *bilinear* pairing of *two* holes living in the Stinespring dilation. In his notation the one-hole vector $R_k(X)$ satisfies $R_k(X)^\dagger R_k(X) = \Phi^2(\Phi(X^\dagger)\Phi(X)) - \Phi(\Phi^2(X^\dagger)\Phi^2(X)) = O(\eta)\|X\|^2$, so $\|R_k(X)\| = O(\sqrt{\eta})\|X\|$, while the defect equals a product $W = V_1^\dagger R_1(X^\dagger)^\dagger u_{3,0}(\Phi(Y)) V_3 R_0(Z)$ of two such vectors, whence $\|W\| = O(\sqrt{\eta}) \cdot O(\sqrt{\eta}) = O(\eta)$ by Cauchy–Schwarz. The two-hole pairing requires complete positivity (it lives in the dilation Hilbert space and uses matrix amplifications $\text{id}_n \otimes \Phi$); a general positive map has no such dilation, so only the single-insertion $O(\sqrt{\eta})$ bound survives. For *decomposable* maps $\Phi = \Phi_0 + \Psi_0 \circ \tau$ (with Φ_0, Ψ_0 completely positive and τ the transpose, with explicit component control such as $\Phi_0(\mathbf{1}) + \Psi_0(\mathbf{1}) = \mathbf{1}$) each summand carries a dilation, and we *conjecture* the two-hole mechanism is restored if those dilation models can be made compatible, giving $\varepsilon = O(\eta)$; this is (OPEN).

Agent B review correction. The decomposable statement is not a theorem from an arbitrary CP+coCP decomposition. A linear η bound would still require compatible dilation bookkeeping or

a direct cancellation argument for the two holes.

8.2 Exact factorization by positive maps

The bridge yields an *approximate* Jordan algebra and approximate factor maps. Producing *exact* unital-positive factor maps requires more: either a concrete genuine Jordan algebra sitting near A inside $B(\mathcal{H})_{\text{sa}}$, or a stability statement promoting the idempotent P to a genuine positive idempotent. The exact endpoint is classical.

Theorem 8.2 (Effros–Størmer; (CITED)). *Let A be a unital JC-algebra and $P: A \rightarrow A$ a unital positive projection. Then $P(A)$, equipped with the given vector-space operations and the product $r \bullet s = P(r \circ s)$, is a JC-algebra. Equivalently, the fixed-point set of any normal unital positive map of a JW-algebra into itself carries a natural JW-algebra structure.*

Theorem 8.2 is the exact template our bridge approximates: it is the Jordan analogue of the Choi–Effros theorem [1] for completely positive projections. If the spectral idempotent P were genuinely positive, Theorem 8.2 would directly furnish a genuine JC-algebra and exact factor maps. The obstruction is that P is only $O(\eta)$ -positive, and one cannot in general round it to a positive map without loss.

Remark 8.3 (Positivity rounding fails generically; (OPEN) stability). A generic dimension-free $O(\varepsilon)$ positivity-rounding lemma is **false**. There is an ε -positive unital map (with spin-factor target $J \cong \mathbb{R} \oplus \mathbb{R}^m$, domain ℓ_∞^{2m} or $B(\mathbb{C}^{2m})_{\text{sa}}$) that is at distance at least $\sqrt{\varepsilon}$ from *every* positive unital map. Consequently an approximate Jordan-isomorphism alone does not yield exact unital-positive factor maps with the bridge exponent. The viable route is *near-positive projection stability*:

$$\begin{aligned} &P \text{ a } \delta\text{-positive unital idempotent, } \|P\| \leq 1 + \delta \\ \implies &\text{dist}(P, \text{positive unital idempotents}) \leq C\sqrt{\delta}, \end{aligned}$$

which, if true, gives a genuine positive idempotent E with $\|E - P\| = O(\sqrt{\eta})$; Theorem 8.2 then applies to E and yields exact unital-positive factor maps through $J = E(B(\mathcal{H})_{\text{sa}})$ with $\|\Delta\Upsilon - \Phi\| = O(\sqrt{\eta})$. This stability statement is (OPEN): it is known classically only at the square-root scale (linear stability is false), and the dimension-free case reduces to an unproved stability theorem for almost-idempotent stochastic matrices.

8.3 Relation to van Luijk–Wilming

The structures here are the approximate counterparts of an exact object from the theory of quantum statistical experiments.

Remark 8.4 (van Luijk–Wilming minimal sufficient Jordan algebra; (CITED)). For a faithful statistical experiment (ρ_θ) , the *minimal sufficient J^* -algebra* is the J^* -algebra

$$J_{(\rho_\theta)} = \bigcap \{ \text{Fix}(T) : T \text{ a } (\rho_\theta)\text{-preserving unital positive map on } L(\mathcal{H}) \},$$

an exact intersection of fixed-point sets of unital positive maps, and it “is generated by the Neyman–Pearson tests” $[\rho > t\sigma]$ [11]. Our near-fixed-point algebra $A = \text{Im } P$ is the *approximate* analogue: P replaces an exact conditional expectation onto a fixed-point set by an approximate idempotent, and $a \bullet b = P(a \circ b)$ replaces the exact Effros–Størmer product $r \bullet s = P(r \circ s)$ of Theorem 8.2. In their setting the Jordan algebra is always special (concretely realised, exceptional $H_3(\mathbb{O})$ excluded); the bridge target inherits this, as recorded in Section 7.

8.4 Status ledger

Table 1 summarises the standing of each component.

Component	Status	Where
Layer-2 bridge (approx. Jordan algebra at $\sqrt{\eta}$)	proved, dimension-free	Section 6
Exponent η (Kitaev strength)	UCP/cb: cited; decomposable: open	Remark 8.1
Exact unital-positive factorization	open (needs projection stability)	Remark 8.3
Layer-1 structure theorem	open (dimension-free homotopy)	Section 7
Exceptional algebras ($H_3(\mathbb{O})$)	channel target special; abstract layer open	Remark 8.4

Table 1: Status of the components of the programme.

The two genuinely open problems are independent. The first is the *exponent*: whether decomposability (a dilation for each CP/co-CP summand) restores the two-hole mechanism and upgrades $\sqrt{\eta}$ to η (Remark 8.1). The second is *exact factorization*: whether a near-positive idempotent is $O(\sqrt{\eta})$ -close to a genuine positive idempotent, after which exact Effros–Størmer applies (Remark 8.3); generic positivity rounding is ruled out by the spin-factor counterexample. The Layer-1 abstract structure theorem (Section 7) is open for a third, orthogonal reason: the dimension-free bounded homotopy that Kitaev obtains from complete positivity has no established Jordan analogue.

References

- [1] Man-Duen Choi and Edward G. Effros. Injectivity and operator spaces. *J. Funct. Anal.*, 24:156–209, 1977.
- [2] Cho-Ho Chu and Bernard Russo. Cohomology of Jordan triples and Lie algebras. *arXiv preprint*, arXiv:1512.03347, 2016.
- [3] Edward G. Effros and Erling Størmer. Positive projections and Jordan structure in operator algebras. *Math. Scand.*, 45:127–138, 1979.
- [4] Jacques Faraut and Adam Korányi. *Analysis on Symmetric Cones*. Oxford University Press, 1994.
- [5] Harald Hanche-Olsen and Erling Størmer. *Jordan Operator Algebras*, volume 21 of *Monographs and Studies in Mathematics*. Pitman, Boston, 1984.
- [6] Pascual Jordan, John von Neumann, and Eugene Wigner. On an algebraic generalization of the quantum mechanical formalism. *Ann. of Math.*, 35:29–64, 1934.
- [7] Richard V. Kadison. A generalized Schwarz inequality and algebraic invariants for operator algebras. *Ann. of Math.*, 56:494–503, 1952.

- [8] Alexei Kitaev. Almost-idempotent quantum channels and approximate c^* -algebras. *arXiv preprint*, arXiv:2405.02434, 2024.
- [9] A. J. Penico. The Wedderburn principal theorem for Jordan algebras. *Trans. Amer. Math. Soc.*, 70:404–420, 1951.
- [10] Erling Størmer. On the Jordan structure of C^* -algebras. *Trans. Amer. Math. Soc.*, 120:438–447, 1965.
- [11] Lauritz van Luijk and Henrik Wilming. Sufficiency and Petz recovery for positive maps. *arXiv preprint*, arXiv:2604.08380, 2026.